

Reachy Mini Integration & Setup Guide

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1. System Overview & Architecture

This system integrates the offline pluggable edge voice assistant (Baymax) running on the **Jetson Orin Nano** with the physical **Reachy Mini robot** controlled by a **Raspberry Pi Zero 2W**. The voice assistant features hybrid offline/online LLM inference, local text-to-speech (Piper), speech-to-text (Whisper), and local knowledge vault (NCERT textbooks / NPS expo info) grounding with DuckDuckGo fallback.

Hardware Communication Ports:

- **Port 5001 (TCP):** Motor Server on Pi Zero (receives joint movement commands and blink configurations).
- **Port 5002 (TCP):** Microphone Server on Pi Zero (streams raw 16kHz audio from physical robot mic).
- **Port 5003 (TCP):** Speaker Server on Pi Zero (receives raw WAV audio streams to play through robot speakers).
- **Port 5000 (TCP):** Camera Server on Pi Zero (streams H264 video feed via rpicam-vid to Jetson).

2. Integrated Code Base Changes

A. Robot Codebase (reachy_mini_custom)

Modified reachy_mini_v6/pi_robot/robot_server.py to execute the rpicam-vid camera streaming subprocess automatically in the background in Python when the server starts, eliminating the need to run run.sh manually:

```
def start_camera_stream():
    def run_camera():
        while True:
            try:
                proc = subprocess.Popen([
                    'rpicam-vid', '-t', '0', '--inline', '--width', '640', '--height', '4
80',
                    '--framerate', '30', '--codec', 'h264', '--listen', '-o', 'tcp://0.0.
0.0:5000'
                ], stdout=subprocess.DEVNULL, stderr=subprocess.DEVNULL)
                proc.wait()
            except Exception as e:
                print(f"Camera stream error/not installed: {e}")
                time.sleep(0.5)
            threading.Thread(target=run_camera, daemon=True).start()

    start_camera_stream()
```

B. Assistant Codebase (small-voice-main)

- **config.yaml & config-orin.yaml:** Appended robot configuration parameters:

```
robot:  
  enabled: false  
  ip: ""
```

- **src/config.py:** Added robot_enabled (bool) and robot_ip (str) to AppConfig with default values for backward compatibility.
- **src/audio/robot.py (New File):** Created `RobotController` that handles socket connections to the Pi Zero. It plays synthesized WAV audio on port 5003 and performs natural neck sway motor gestures on port 5001 during speech.
- **src/main.py:** Integrated `RobotController`. If robot.enabled is set to true, the local audio stream is bypassed and sent directly to the robot speaker server.

3. Wake Word Training & Integration

To replace the old Jarvis wake word with **Baymax**, a custom classification model was trained and integrated:

- **scripts/train_wakeword.py (New File):** Automates training. Synthesizes positive clips ('Hey Baymax', 'Baymax') and negative clips ('Hey Jarvis', 'Alexa', 'Siri', etc.) using your local Piper TTS engine, varies speeds and volume, extracts features, trains a PyTorch Classifier, and exports the final model to ONNX.
- **assets/wakeword_models/hey_baymax_v0.1.onnx & .data (New Files):** Pre-trained custom ONNX wake word classification model and weight files, generated by the training script.
- **src/audio/wakeword.py:** Updated default wake word model path to the custom hey_baymax_v0.1.onnx model and set detection sensitivity threshold to 0.2.
- **src/main.py:** Updated local mic loop initializer to load the hey_baymax_v0.1.onnx model.

4. Step-by-Step Setup & Execution Guide

Follow these steps to run the complete setup from scratch:

Step 1: Start the Robot Server (on the Pi Zero 2W)

1. SSH into the Pi Zero 2W.
2. Navigate to the Pi robot code directory:
`cd /rechy_mini/Rechy_mini_custom/rechy_mini_v6/pi_robot`
3. Start the server:
`python3 robot_server.py`

Step 2: Configure the Assistant (on Jetson Orin Nano / Windows)

1. Open config.yaml (or copy config-orin.yaml to config.yaml on the Jetson).
2. Enable the robot connection and enter the Pi Zero's IP address:
robot:
 enabled: true
 ip: "192.168.1.XX" # <-- Put your Pi Zero's IP here

Step 3: Run the Main Assistant loop

1. Open a terminal in the small-voice-main directory.
2. Activate your virtual environment (e.g. source jarvis-env/bin/activate on Jetson).
3. Execute the assistant:
`python3 src/main.py`